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Laboratory Work 2:

Criptanaliza cifrurilor monoalfabetice

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1 Introducere și Obiective

Cifrurile cu substituție monoalfabetică sunt printre cele mai simple și vechi metode de ascundere a unui mesaj. Principiul de funcționare este direct: fiecare literă din alfabetul clar este înlocuită constant cu o altă literă sau simbol pe tot parcursul textului.

Deși spațiul teoretic al cheilor posibile este imens ($26! \approx 4{,}03 \times 10^{26}$ permutări, ceea ce face imposibilă o căutare manuală prin forță brută), cifrul are o vulnerabilitate majoră: păstrează intacte frecvențele caracteristice ale literelor din limba în care a fost redactat textul. Dacă mesajul este scris în engleză, literele și grupurile de litere cele mai frecvente (cum ar fi vocala `e`, consoana `t` sau cuvântul `the`) se vor reflecta clar în textul cifrat, permițând spargerea lui fără a cunoaște cheia inițială.

Obiectivele acestei lucrări de laborator sunt:

1. Înțelegerea principiului analizei de frecvență ca tehnică clasică de criptanaliză împotriva cifrurilor monoalfabetice.

2. Determinarea frecvențelor literelor din criptograma primită (Varianta 22) și compararea lor cu distribuția standard a limbii engleze.

3. Utilizarea utilitarului online indicat în îndrumar (`https://crypto.interactive-maths.com/`) pentru a asista procesul de substituție.

4. Descifrarea progresivă a mesajului interceptat, urmărind metodologia din secțiunea 2.3 a îndrumarului și afișând starea textului la fiecare iterație.

5. Reconstituirea alfabetului cheii de criptare sub forma Tabelului 2.5 din îndrumar.

2 Noțiuni Teoretice și Instrumentul de Lucru

2.1 Principiul analizei de frecvență

Punctul slab al oricărui cifru monoalfabetic este că nu ascunde frecvența de apariție a caracterelor. Dacă avem un text cifrat suficient de lung și cunoaștem limba în care a fost redactat mesajul, putem deduce o mare parte din litere doar analizând cât de des apar. Pe măsură ce lungimea mesajului crește, procentele literelor din cifru tind să se apropie de cele din limba naturală respectivă.

Pentru limba engleză, distribuția frecvențelor (conform Tabelului 2.2 din îndrumarul de laborator) arată că litera `E` este de departe cea mai des întâlnită (aproximativ 12,70%), urmată de litere precum `T`, `A`, `O`, `I`, `N`, `S`, `H` și `R`. La polul opus, litere precum `J`, `X`, `Q` și `Z` apar extrem de rar (sub 0,2%).

Table 1: Frecvența etalon a literelor limbii engleze (Tabelul 2.2 din îndrumar).

Lit.	**%**	**Lit.**	**%**	**Lit.**	**%**	**Lit.**	**%**	**Lit.**	**%**	**Lit.**	**%**	**Lit.**	**%**
A	8,17	E	12,70	I	6,97	M	2,41	Q	0,09	U	2,76	Y	1,97
B	1,49	F	2,23	J	0,15	N	6,75	R	5,99	V	0,98	Z	0,07
C	2,78	G	2,01	K	0,77	O	7,51	S	6,33	W	2,36	–	–
D	4,25	H	6,09	L	4,03	P	1,93	T	9,06	X	0,15	–	–

2.2 Metodologia criptanalizei monoalfabetice

O simplă ordonare descrescătoare urmată de o substituție mecanică directă conform frecvențelor globale nu oferă întotdeauna rezultatul corect, deoarece fiecare text are propriile variații de vocabular. De aceea, procesul implică o analiză logică, combinând frecvențele literelor cu alte repere utile:

- **Digrafe frecvente:** Perechi de litere des întâlnite, cum ar fi `TH`, `HE`, `IN`, `ER`, `AN`, `RE`, `ED`, `ON`, `ES`, `ST`, `EN`, `AT`.
- **Trigrafe frecvente:** Grupuri de 3 litere foarte comune, în special `THE`, `AND`, `THA`, `ENT`, `ION`, `TIO`, `FOR`.
- **Cuvinte scurte:** În engleză există doar două cuvinte uzuale dintr-o singură literă: `"A"` și `"I"`. De asemenea, cuvintele de două litere sunt foarte comune și ușor de recunoscut în context: `"TO"`, `"OF"`, `"IN"`, `"IS"`, `"IT"`, `"HE"`, `"WE"`, `"AS"`, `"AT"`, `"BE"`.
- **Litere duble:** Litere care apar des repetate: `SS`, `EE`, `TT`, `FF`, `LL`, `OO`.

2.3 Instrumentul utilizat pentru analiză și substituție

Conform indicațiilor din lucrarea de laborator, criptanaliza a fost efectuată folosind aplicația web interactivă:

https://crypto.interactive-maths.com/frequency-analysis-breaking-the-code.html

Modul de lucru în cadrul acestui utilitar a fost următorul:

1. **Câmpul de intrare (Intercept):** Am încărcat textul cifrat brut. Serviciul convertește automat literele în majuscule și păstrează neschimbate spațiile și semnele de punctuație.

2. **Modulul statistic (Find Frequencies):** La apăsarea butonului *"Find Frequencies"*, aplicația a determinat aparițiile fiecărei litere și procentul asociat, ordonându-le descrescător în paralel cu distribuția etalon a limbii engleze.

3. **Identificarea secvențelor repetate (Count Digraphs, Trigraphs, Doubles):** Opțiunile dedicate au permis găsirea rapidă a celor mai frecvente digrafe, trigrafe și litere duble din text.

4. **Mecanismul de substituție (Make Substitutions):** Sub fiecare literă cifrată din tabel există o căsuță unde am completat litera clară dedusă (cu literă mică). La apăsarea butonului *"Make Substitutions"*, textul s-a actualizat instantaneu: literele înlocuite apar cu caractere mici, iar literele încă nedescifrate rămân cu majuscule, permițând urmărirea progresivă a fiecărui pas.

3 Desfășurarea procesului de spargere a criptogramei

Fie că am interceptat o criptogramă *c* (Varianta 22), despre care cunoaștem că a fost obținută în urma utilizării unui cifru monoalfabetic peste un mesaj scris în limba engleză (caracterele non-alfabetice rămânând necriptate):

```
RQXSV QVIAVIW FTIOSVF ZTF AV WQV AVPW LGNRG HIFUWNSNJXPW, DGHNGWVPWTASF WQV
JIVTWVPW XP RXSSXTZ CIVOVIXHL CIXVOZTG. DGSXLV QXPHNGWVZUNITIF, QXP
VZXGVGHV XP ODV ZNPW VZUQTWXHTSSF WN RQTW QV OXO. XGOVVO, WRN ZNIV
OXPPXZXSTI ZVG XG T PXGJSV CXVSO HTG PHTIHVSF AVXZTJXGVO. RQVIV FTIOSVF RTP
ITAVSTXPXTG, NDWJNXGJ, PDUVICXHXTS, CIVV TGOVTPF RXWQ WQV OVWTXSP NC T JNNO
```

```
PWNIF, TGO VKVI IVTOF CNI WQV ZTXG HQTGHV, CIXVOZTG WVGOVO WNRTIO
XGWINKVIPXNG, OVUWQ NC PWDOF, UVIPNGTS PVHDIXWF, WXZXOXWF, OVOXHTWXNG, TGO
THHDITHF, GXHVWF, TGO KTSXOXWF NC RNIL. OVPUXWVWQV IVSTWXKV OITAGVPP NC
WQVPV UVIPNGTS WITXWP--NI UVIQTUP AVHTDPV NCWQVZ, CIXVOZTG'P WQVNIVWXHTS
HNGWIXADWXNGP TGO QXP UITHWXHTS TWWTXGZVGWPVYHVVO WQNPV NC TGF NWQVI
HIFUWNSNJXPW. FTIOSVF'P HTIVVI RTP SXLV TGTZTMXGJ PLFINHLVW WQTW VYUSNOVP
XG CTGWTPWXH UTWWVIGP TJTXGPW WQVQVTKVGP. CIXVOZTG'P RTP SXLV WQV PDG. QV
RTP ANIG RNSCV CIXVOZTG NG PVUWVZAVI 24, 1891, XG LXPQXGVK, IDPPXT, WQV
NSOVPW PNG TGO PVHNGO HQXSO NC CIVOVIXHL TGO INPT CIXVOZTG. QXP CTWQVI, T
IDZTGXTG RQN PUNLV VXJQW STGJDTJVP TGO RNILVO TP TGXGWVIUIVWVI CNI WQV
IDPPXTG UNPW NCCXHV, VZXJITWVO WN TZVIXHT XG 1892, TW RQXHQ WXZV QXP PNG'P
GTZV RTP HQTGJVO WN RXSSXTZ. WQV CTZXSF PVWWSVOXG UXWWPADIJQ, RQVIV QXP
CTWQVI ZTGTJVO T PVRXGJ ZTHQXGV TJVGHF. RXSSXTZ JITODTWVO XG 1909 TP NGV NC
WQV WVG QNGNI PWDOVGWP XG T HSTPP NC300 TW UXWWPADIJQ HVGWITS QXJQ PHQNNS;
QV WQVG RVGW WN RNIL TP HQXVCHSVIL XG WQV VIXV HXWF XING RNILP, T CXIZ WQTW
PNSO PWVTZ VGJXGVP. TANDWWQTW WXZV WQV ATHL-WN-WQV-CTIZ ZNKVZVGW HTSSVO WN
HXWF ANFP, TGO XG WQVCTSS NC 1910, CIXVOZTG TGO WQIVV CIXVGOP VGINSSVO XG
ZXHQXJTGTJIXHDSWDITS HNSSVJV, RQNPV HQXVC TWWITHWXNG RTP WQTW XW RTP
WDXWXNG-CIVV. ADW CIXVOZTG PNNG OXPHNKVIVO WQTW CTIZXGJ QVSO SXWWSV
XGWVIVPW CNI QXZ. QV RTP TG XGKVGWXKV FNDGJ CVSSNR RQN SXLVO WN CXY WQXGJP
TGO QTO RIXWWVGPNZV PHXVGHV CXHWXNG CNI QXP QXJQ-PHQNNS UTUVI; QV RTP
ITUXOSF HNZXGJ WNWQV HNGHSDPXNG WQTW QV SXLVO PHXVGHV. TW WQV VGO NC WQV
WVIZ QV SVTIGVOWQTW WDXWXNG RTP TSPN CIVV XG T PHXVGWXCXH CXVSO TSSXVO WN
TJIXHDSWDIV--JVGVWXHP--TW NGV NC WQV XKF SVTJDV DGXKVIPXWXVP, HNIGVSS. QV
ANIINRVOWITXG CTIV TGO TIIXKVO XG XWQTHT, GVR FNIL, XG CVAIDTIF, 1911,
RQVIV QVJNW T ENA RTXWXGJ NG WTASVP. TCWVI HNZZVGHVZVGW XG CVAIDTIF NC 1914,
QVTWWVGOVO JITODTWV PHQNNS, ZTGTJXGJ WN CTSS XG SNKV WRXHV, NGHV RXWQ
TAIDGVWWV, NGHV RXWQ WQV ASNGOV OTDJQWVI NC T ZNKXV-QNDPV NRGVI. RQXSVQV
RTP WQVIV, T RVTSWQF WVYWXSV ZVIHQTGW, JVNIJV CTAFTG, RQNZTXGWTXGVO
STANITWNIXVP XG THNDPWXHP, HQVZXPWIF, JVGVWXHP, TGO HIFU-WNSNJF (WN WIF WN
UINKV WQTW ATHNG RINWV PQTLVPUVTIV'P USTFP) NG QXP 500-THIV VPWTWV,
IXKVIATGL, TW JVGVKT, XSSXGNXP, OVHXOVO WQTW QV GVVOVO TJVGVWXHXPW WN
XZUINKV WQV JITXGP TGO SXKVPWNHL NG QXP CTIZ. QV TUUSXVO WNHNIGVSS CNI T
"RNDSO-AV-VI," GNW TG "TP-XP-VI," TGO QXIVO CIXVOZTG, WNAVJXG EDGV 1, 1915.
CTAFTG RTP T ZTG NC GN CNIZTS VODHTWXNG ADW NC XGWVSSXJVGHV TGOVGVIJF. QV
QTO T JIVTW OVPXIV WN AV "PNZVANOF," TGO WQTW OVPXIVZNWXKTWVO QXP
PDAPXOXMXGJ WQV ATHNGXTG PWDOXVP: UINNC NC WQXPIVKNSDWXNGTIF WQVPXP RNDSO
HNKVI XWP UTWING TP RVSS TP XWP THWDTSOXPHNKVIVIP RXWQ JSNIF. QV QXZPVSC
IVTO SXWWSV, ADW QV TAPNIAVO VGNDJQCINZ WQNPV TINDGO QXZ WN ZTLV QXP WTSL
NG TSZNPW TGF PDAEVHW PNDGOXZUIVPPXKV--TW SVTPW PDUVICXHXTSSF. QV RTP
TDWNHITWXH, GVKVI TSSNRXGJ QXPPWTCC WN OXPTJIVV RXWQ QXZ, ADW NWQVIRXPV GNW
DGUSVTPTGW PN SNGJ TPVZUSNFVVP IVHNJGXMVO WQTW QV RTP ANPP. T HTIOXGTS
TIWXHSV NC CTXWQ RXWQQXZ RTP WQTW T RVSS-VYVHDWVO PTSVP HTZUTXJG HNDSO UDW
THINPP TSZNPWTGFWQXGJ.
```

Primul pas este să găsim frecvențele tuturor literelor care apar în criptogramă, așa cum e arătat în tabelul 2.3. Pentru aceasta, am introdus mesajul în câmpul *"Intercept"* și am apăsat *"Find Frequencies"*.

Table 2: Frecvența literelor în criptograma interceptată (Tabelul 2.3).

Ciphertext Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
Frequency	42	0	74	59	3	51	197	102	171	60	26	24	3
Ciphertext Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	189	116	176	140	60	120	260	38	346	249	234	5	72

Pentru comoditate ordonăm tabelul în descreștere a frecvențelor, așa cum e arătat în tabelul

2.4.

Table 3: Frecvența literelor în criptograma interceptată ordonată descrescător (Tabelul 2.4).

Ciphertext Letter	V	T	W	X	G	N	P	I	Q	S	O	H	C
Frequency	346	260	249	234	197	189	176	171	140	120	116	102	74
Ciphertext Letter	Z	J	R	D	F	A	U	K	L	Y	E	M	B
Frequency	72	60	60	59	51	42	38	26	24	5	3	3	0

Mesajul interceptat are în total 2,817 litere alfabetice. Comparând datele din tabelul 2.4 cu frecvențele teoretice din tabelul 2.2, putem începe să deducem primele litere.

3.1 Iterația 1: Determinarea nucleului `THE`

Acum că avem frecvențele literelor din textul cifrat, putem începe să facem primele substituții. Vedem că cea mai frecventă literă din textul cifrat este `V` (cu 346 de apariții, reprezentând 12,28%), urmată de `T` (260) și `W` (249). Comparând cu tabelul 2.2, unde `E` este de departe cea mai comună literă (12,70%), putem deduce cu certitudine că litera `V` reprezintă vocala `e`.

Mai mult, folosind butonul *"Count Trigraphs"*, observăm că trigraful `WQV` este de departe cel mai des întâlnit în tot mesajul: apare de 52 de ori în total și de 37 de ori ca un cuvânt separat de trei litere. În limba engleză cel mai comun cuvânt de 3 litere este `"the"`, ceea ce se potrivește perfect cu faptul că ultima literă `V` este deja dedusă ca `e`. Deducem așadar:

$$\text{"W"} \to \text{"t"}, \quad \text{"Q"} \to \text{"h"}, \quad \text{"V"} \to \text{"e"}.$$

Am introdus `t`, `h` și `e` în căsuțele corespunzătoare din dreptul literelor `W`, `Q` și `V` din tabelul aplicației și am apăsat butonul *"Make Substitutions"*, obținând:

```
RhXSe heIAeIt FTIOSeF ZTF Ae the AePt LGNRG HIFUtNSNJXPt, DGHNGtePtTASF the
JIeTtePt XP RXSSXTZ CIeOeIXHL CIXeOZTG. DGSXLe hXPHNGteZUNITIF, hXP
eZXGeGHe XP ODe ZNPt eZUhTtXHTSSF tN RhTt he OXO. XGOeeO, tRN ZNIe
OXPPXZXSTI ZeG XG T PXGJSe CXeSO HTG PHTIHeSF AeXZTJXGeO. RheIe FTIOSeF RTP
ITAeSTXPXTG, NDtJNXGJ, PDUeICXHXTS, CIee TGOeTPF RXth the OetTXSP NC T JNNO
PtNIF, TGO eKeI IeTOF CNI the ZTXG HhTGHe, CIXeOZTG teGOeO tNRTIO
XGtINKeIPXNG, OeUth NC PtDOF, UeIPNGTS PeHDIXtF, tXZXOXtF, OeOXHTtXNG, TGO
THHDITHF, GXHetF, TGO KTSXOXtF NC RNIL. OePUXtethe IeSTtXKe OITAGePP NC
thePe UeIPNGTS tITXtP--NI UeIhTUP AeHTDPe NCtheZ, CIXeOZTG'P theNIetXHTS
HNGtIXADtXNGP TGO hXP UITHtXHTS TttTXGZeGtPeYHeeO thNPe NC TGF NtheI
HIFUtNSNJXPt.  FTIOSeF'P HTIeeI RTP SXLe TGTZTMXGJ PLFINHLet thTt eYUSNOeP
XG CTGtTPtXH UTtteIGP TJTXGPt theheTKeGP. CIXeOZTG'P RTP SXLe the PDG. he
RTP ANIG RNSCe CIXeOZTG NG PeUteZAeI 24, 1891, XG LXPhXGeK, IDPPXT, the
NSOePt PNG TGO PeHNGO HhXSO NC CIeOeIXHL TGO INPT CIXeOZTG. hXP CTtheI, T
IDZTGXTG RhN PUNLe eXJht STGJDTJeP TGO RNILeO TP TGXGteIUIeteI CNI the
IDPPXTG UNPt NCCXHe, eZXJITteO tN TZeIXHT XG 1892, Tt RhXHh tXZe hXP PNG'P
GTZe RTP HhTGJeO tN RXSSXTZ. the CTZXSF PettSeOXG UXttPADIJh, RheIe hXP
CTtheI ZTGTJeO T PeRXGJ ZTHhXGe TJeGHF. RXSSXTZ JITODTteO XG 1909 TP NGe NC
the teG hNGNI PtDOeGtP XG T HSTPP NC300 Tt UXttPADIJh HeGtITS hXJh PHhNNS;
he theG ReGt tN RNIL TP HhXeCHSeIL XG the eIXe HXtF XING RNILP, T CXIZ thTt
PNSO PteTZ eGJXGeP. TANDtthTt tXZe the ATHL-tN-the-CTIZ ZNKeZeGt HTSSeO tN
HXtF ANFP, TGO XG theCTSS NC 1910, CIXeOZTG TGO thIee CIXeGOP eGINSSeO XG
ZXHhXJTGTJIXHDStDITS HNSSeJe, RhNPe HhXeC TttITHtXNG RTP thTt Xt RTP
tDXtXNG-CIee.  ADt CIXeOZTG PNNG OXPHNKeIeO thTt CTIZXGJ heSO SXttSe
XGteIePt CNI hXZ. he RTP TG XGKeGtXKe FNDGJ CeSSNR RhN SXLeO tN CXY thXGJP
TGO hTO RIXtteGPNZe PHXeGHe CXHtXNG CNI hXP hXJh-PHhNNS UTUeI; he RTP
ITUXOSF HNZXGJ tNthe HNGHSDPXNG thTt he SXLeO PHXeGHe.  Tt the eGO NC the
```

```
teIZ he SeTIGeOthTt tDXtXNG RTP TSPN CIee XG T PHXeGtXCXH CXeSO TSSXeO tN
TJIXHDStDIe--JeGetXHP--Tt NGe NC the XKF SeTJDe DGXKeIPXtXeP, HNIGeSS. he
ANIINReOtITXG CTIe TGO TIIXKeO XG XthTHT, GeR FNIL, XG CeAIDTIF, 1911,
RheIe heJNt T ENA RTXtXGJ NG tTASeP. TCteI HNZZeGHeZeGt XG CeAIDTIF NC 1914,
heTtteGOeO JITODTte PHhNNS, ZTGTJXGJ tN CTSS XG SNKe tRXHe, NGHe RXth
TAIDGette, NGHe RXth the ASNGOe OTDJhteI NC T ZNKXe-hNDPe NRGeI. RhXSehe
RTP theIe, T ReTSthF teYtXSe ZeIHhTGt, JeNIJe CTAFTG, RhNZTXGtTXGeO
STANITtNIXeP XG THNDPtXHP, HheZXPtIF, JeGetXHP, TGO HIFU-tNSNJF (tN tIF tN
UINKe thTt ATHNG RINte PhTLePUeTIe'P USTFP) NG hXP 500-THIe ePtTte,
IXKeIATGL, Tt JeGeKT, XSSXGNXP, OeHXOeO thTt he GeeOeO TJeGetXHXPt tN
XZUINKe the JITXGP TGO SXKePtNHL NG hXP CTIZ. he TUUSXeO tNHNIGeSS CNI T
"RNDSO-Ae-eI," GNt TG "TP-XP-eI," TGO hXIeO CIXeOZTG, tNAeJXG EDGe 1, 1915.
CTAFTG RTP T ZTG NC GN CNIZTS eODHTtXNG ADt NC XGteSSXJeGHe TGOeGeIJF. he
hTO T JIeTt OePXIe tN Ae "PNZeANOF," TGO thTt OePXIeZNtXKTteO hXP
PDAPXOXMXGJ the ATHNGXTG PtDOXeP: UINNC NC thXPIeKNSDtXNGTIF thePXP RNDSO
HNKeI XtP UTtING TP ReSS TP XtP THtDTSOXPHNKeIeIP RXth JSNIF. he hXZPeSC
IeTO SXttSe, ADt he TAPNIAeO eGNDJhCINZ thNPe TINDGO hXZ tN ZTLe hXP tTSL
NG TSZNPt TGF PDAEeHt PNDGOXZUIePPXKe--Tt SeTPt PDUeICXHXTSSF. he RTP
TDtNHITtXH, GeKeI TSSNRXGJ hXPPtTCC tN OXPTJIee RXth hXZ, ADt NtheIRXPe GNt
DGUSeTPTGt PN SNGJ TPeZUSNFeeP IeHNJGXMeO thTt he RTP ANPP. T HTIOXGTS
TItXHSe NC CTXth RXthhXZ RTP thTt T ReSS-eYeHDteO PTSeP HTZUTXJG HNDSO UDt
THINPP TSZNPtTGFthXGJ.
```

3.2 Iterația 2: Descifrarea cuvintelor funcționale scurte (`to`, `that`, `what`, `is`)

Privind textul parțial obținut după prima iterație, încep să se contureze câteva cuvinte scurte foarte comune:

1. Secvența `wn` apare foarte des ca un cuvânt separat de două litere (28 de apariții), având forma `"tN"`. În engleză, cea mai comună prepoziție scurtă care începe cu `t` este `"to"`, ceea ce ne arată că `N` este `o`.

2. Cuvântul cifrat `wqtw` are acum forma `"thTt"`. În limba engleză acesta este în mod clar cuvântul `"that"`, ceea ce înseamnă că `T` este `a`.

3. În sintagma `"tN RhTt he OXO"` → `"to Rhat he OXO"`, cuvântul `rqtw` are forma `"Rhat"`. Din context este evident vorba despre `"what"`, stabilind că `R` reprezintă `w`.

4. În propoziția inițială `"the JIeTtePt XP RXSSXTZ"`, observăm cuvântul de două litere `xp`. După superlativul `"the greatest"` este firesc să urmeze verbul copulativ `"is"` înaintea numelui de persoană. Această deducție ne oferă corespondențele `X` → `i` și `P` → `s`.

Am introdus aceste litere în instrument (`N` → `o`, `T` → `a`, `R` → `w`, `X` → `i`, `P` → `s`) și am apăsat *"Make Substitutions"*:

```
whiSe heIAeIt FaIOSeF ZaF Ae the Aest LGowG HIFUtoSoJist, DGHoGtestaASF the
JIeatest is wiSSiaZ CIeOeIiHL CIieOZaG. DGSiLe hisHoGteZUoIaIF, his
eZiGeGHe is ODe Zost eZUhatiHaSSF to what he OiO. iGOeeO, two ZoIe
OissiZiSaI ZeG iG a siGJSe CieSO HaG sHaIHeSF AeiZaJiGeO. wheIe FaIOSeF was
IaAeSaisiaG, oDtJoiGJ, sDUeICiHiaS, CIee aGOeasF with the OetaiSs oC a JooO
stoIF, aGO eKeI IeaOF CoI the ZaiG HhaGHe, CIieOZaG teGOeO towaIO
iGtIoKeIsioG, OeUth oC stDOF, UeIsoGaS seHDIitF, tiZiOitF, OeOiHatioG, aGO
aHHDIaHF, GiHetF, aGO KaSiOitF oC woIL. OesUitethe IeSatiKe OIaAGess oC
these UeIsoGaS tIaits--oI UeIhaUs AeHaDse oCtheZ, CIieOZaG's theoIetiHaS
HoGtIiADtioGs aGO his UIaHtiHaS attaiGZeGtseYHeeO those oC aGF otheI
HIFUtoSoJist.  FaIOSeF's HaIeeI was SiLe aGaZaMiGJ sLFIoHLet that eYUSoOes
```

```
iG CaGtastiH UatteIGs aJaiGst theheaKeGs.  CIieOZaG's was SiLe the sDG. he
was AoIG woSCe CIieOZaG oG seUteZAeI 24, 1891, iG LishiGeK, IDssia, the
oSOest soG aGO seHoGO HhiSO oC CIeOeIiHL aGO Iosa CIieOZaG. his CatheI, a
IDZaGiaG who sUoLe eiJht SaGJDaJes aGO woILeO as aGiGteIUIeteI CoI the
IDssiaG Uost oCCiHe, eZiJIateO to aZeIiHa iG 1892, at whiHh tiZe his soG's
GaZe was HhaGJeO to wiSSiaZ. the CaZiSF settSeOiG UittsADIJh, wheIe his
CatheI ZaGaJeO a sewiGJ ZaHhiGe aJeGHF. wiSSiaZ JIaODateO iG 1909 as oGe oC
the teG hoGoI stDOeGts iG a HSass oC300 at UittsADIJh HeGtIaS hiJh sHhooS;
he theG weGt to woIL as HhieCHSeIL iG the eIie HitF iIoG woILs, a CiIZ that
soSO steaZ eGJiGes.  aAoDtthat tiZe the AaHL-to-the-CaIZ ZoKeZeGt HaSSeO to
HitF AoFs, aGO iG theCaSS oC 1910, CIieOZaG aGO thIee CIieGOs eGIoSSeO iG
ZiHhiJaGaJIiHDStDIaS HoSSeJe, whose HhieC attIaHtioG was that it was
tDitioG-CIee.  ADt CIieOZaG sooG OisHoKeIeO that CaIZiGJ heSO SittSe
iGteIest CoI hiZ. he was aG iGKeGtiKe FoDGJ CeSSow who SiLeO to CiY thiGJs
aGO haO wIitteGsoZe sHieGHe CiHtioG CoI his hiJh-sHhooS UaUeI; he was
IaUiOSF HoZiGJ tothe HoGHSDsioG that he SiLeO sHieGHe.  at the eGO oC the
teIZ he SeaIGeOthat tDitioG was aSso CIee iG a sHieGtiCiH CieSO aSSieO to
aJIiHDStDIe--JeGetiHs--at oGe oC the iKF SeaJDe DGiKeIsities, HoIGeSS. he
AoIIoweOtIaiG CaIe aGO aIIiKeO iG ithaHa, Gew FoIL, iG CeAIDaIF, 1911,
wheIe heJot a EoA waitiGJ oG taASes.  aCteI HoZZeGHeZeGt iG CeAIDaIF oC
1914, heatteGOeO JIaODate sHhooS, ZaGaJiGJ to CaSS iG SoKe twiHe, oGHe with
aAIDGette, oGHe with the ASoGOe OaDJhteI oC a ZoKie-hoDse owGeI. whiSehe
was theIe, a weaSthF teYtiSe ZeIHhaGt, JeoIJe CaAFaG, whoZaiGtaiGeO
SaAoIatoIies iG aHoDstiHs, HheZistIF, JeGetiHs, aGO HIFU-toSoJF (to tIF to
UIoKe that AaHoG wIote shaLesUeaIe's USaFs) oG his 500-aHIe estate,
IiKeIAaGL, at JeGeKa, iSSiGois, OeHiOeO that he GeeOeO aJeGetiHist to
iZUIoKe the JIaiGs aGO SiKestoHL oG his CaIZ. he aUUSieO toHoIGeSS CoI a
"woDSO-Ae-eI," Got aG "as-is-eI," aGO hiIeO CIieOZaG, toAeJiG EDGe 1, 1915.
CaAFaG was a ZaG oC Go CoIZaS eODHatioG ADt oC iGteSSiJeGHe aGOeGeIJF. he
haO a JIeat OesiIe to Ae "soZeAoOF," aGO that OesiIeZotiKateO his
sDAsiOiMiGJ the AaHoGiaG stDOies:  UIooC oC thisIeKoSDtioGaIF thesis woDSO
HoKeI its UatIoG as weSS as its aHtDaSOisHoKeIeIs with JSoIF. he hiZseSC
IeaO SittSe, ADt he aAsoIAeO eGoDJhCIoZ those aIoDGO hiZ to ZaLe his taSL
oG aSZost aGF sDAEeHt soDGOiZUIessiKe--at Seast sDUeICiHiaSSF. he was
aDtoHIatiH, GeKeI aSSowiGJ hisstaCC to OisaJIee with hiZ, ADt otheIwise Got
DGUSeasaGt so SoGJ aseZUSoFees IeHoJGiMeO that he was Aoss.  a HaIOiGaS
aItiHSe oC Caith withhiZ was that a weSS-eYeHDteO saSes HaZUaiJG HoDSO UDt
aHIoss aSZostaGFthiGJ.
```

3.3 Iterația 3: Identificarea conjuncțiilor și expresiilor uzuale (`in`, `and`, `may be the best`)

Analizând noul pasaj, recunoaștem ușor alte câteva cuvinte uzuale:

1. În secvența `"iG a siGJSe CieS"`, cuvântul de două litere `"iG"` dinaintea lui `"a"` este clar prepoziția `"in"`, deci `G` este `n`.

2. Cuvântul cifrat `tgo` apare de 17 ori și are forma `"anO"`. În engleză este conjuncția copulativă de bază `"and"`, ceea ce confirmă că `O` este `d`.

3. Chiar în prima frază a textului avem structura `"ZaF Ae the Aest"`. Luând în considerare literele deja decriptate, structura seamănă izbitor cu expresia colocvială `"may be the best"`, ceea ce ne oferă simultan trei litere cheie: `Z` $\rightarrow$ `m`, `F` $\rightarrow$ `y` și `A` $\rightarrow$ `b`.

   Am aplicat aceste substituții în instrument și am acționat *"Make Substitutions"*, obținând:

```
whiSe heIbeIt yaIdSey may be the best Lnown HIyUtoSoJist, DnHontestabSy the
JIeatest is wiSSiam CIedeIiHL CIiedman.  DnSiLe hisHontemUoIaIy, his
```

```
eminenHe is dDe most emUhatiHaSSy to what he did.  indeed, two moIe
dissimiSaI men in a sinJSe CieSd Han sHaIHeSy beimaJined.  wheIe yaIdSey
was IabeSaisian, oDtJoinJ, sDUeICiHiaS, CIee andeasy with the detaiSs oC a
Jood stoIy, and eKeI Ieady CoI the main HhanHe, CIiedman tended towaId
intIoKeIsion, deUth oC stDdy, UeIsonaS seHDIity, timidity, dediHation, and
aHHDIaHy, niHety, and KaSidity oC woIL. desUitethe IeSatiKe dIabness oC
these UeIsonaS tIaits--oI UeIhaUs beHaDse oCthem, CIiedman's theoIetiHaS
HontIibDtions and his UIaHtiHaS attainmentseYHeed those oC any otheI
HIyUtoSoJist.  yaIdSey's HaIeeI was SiLe anamaMinJ sLyIoHLet that eYUSodes
in CantastiH UatteIns aJainst theheaKens.  CIiedman's was SiLe the sDn.  he
was boIn woSCe CIiedman on seUtembeI 24, 1891, in LishineK, IDssia, the
oSdest son and seHond HhiSd oC CIedeIiHL and Iosa CIiedman.  his CatheI, a
IDmanian who sUoLe eiJht SanJDaJes and woILed as aninteIUIeteI CoI the
IDssian Uost oCCiHe, emiJIated to ameIiHa in 1892, at whiHh time his son's
name was HhanJed to wiSSiam.  the CamiSy settSedin UittsbDIJh, wheIe his
CatheI manaJed a sewinJ maHhine aJenHy.  wiSSiam JIadDated in 1909 as one
oC the ten honoI stDdents in a HSass oC300 at UittsbDIJh HentIaS hiJh
sHhooS; he then went to woIL as HhieCHSeIL in the eIie Hity iIon woILs, a
CiIm that soSd steam enJines.  aboDtthat time the baHL-to-the-CaIm moKement
HaSSed to Hity boys, and in theCaSS oC 1910, CIiedman and thIee CIiends
enIoSSed in miHhiJanaJIiHDStDIaS HoSSeJe, whose HhieC attIaHtion was that
it was tDition-CIee.  bDt CIiedman soon disHoKeIed that CaIminJ heSd SittSe
inteIest CoI him.  he was an inKentiKe yoDnJ CeSSow who SiLed to CiY thinJs
and had wIittensome sHienHe CiHtion CoI his hiJh-sHhooS UaUeI; he was
IaUidSy HominJ tothe HonHSDsion that he SiLed sHienHe.  at the end oC the
teIm he SeaInedthat tDition was aSso CIee in a sHientiCiH CieSd aSSied to
aJIiHDStDIe--JenetiHs--at one oC the iKy SeaJDe DniKeIsities, HoIneSS. he
boIIowedtIain CaIe and aIIiKed in ithaHa, new yoIL, in CebIDaIy, 1911,
wheIe heJot a Eob waitinJ on tabSes.  aCteI HommenHement in CebIDaIy oC
1914, heattended JIadDate sHhooS, manaJinJ to CaSS in SoKe twiHe, onHe with
abIDnette, onHe with the bSonde daDJhteI oC a moKie-hoDse owneI. whiSehe
was theIe, a weaSthy teYtiSe meIHhant, JeoIJe Cabyan, whomaintained
SaboIatoIies in aHoDstiHs, HhemistIy, JenetiHs, and HIyU-toSoJy (to tIy to
UIoKe that baHon wIote shaLesUeaIe's USays) on his 500-aHIe estate,
IiKeIbanL, at JeneKa, iSSinois, deHided that he needed aJenetiHist to
imUIoKe the JIains and SiKestoHL on his CaIm.  he aUUSied toHoIneSS CoI a
"woDSd-be-eI," not an "as-is-eI," and hiIed CIiedman, tobeJin EDne 1, 1915.
Cabyan was a man oC no CoImaS edDHation bDt oC inteSSiJenHe andeneIJy.  he
had a JIeat desiIe to be "somebody," and that desiIemotiKated his
sDbsidiMinJ the baHonian stDdies:  UIooC oC thisIeKoSDtionaIy thesis woDSd
HoKeI its UatIon as weSS as its aHtDaSdisHoKeIeIs with JSoIy.  he himseSC
Iead SittSe, bDt he absoIbed enoDJhCIom those aIoDnd him to maLe his taSL
on aSmost any sDbEeHt soDndimUIessiKe--at Seast sDUeICiHiaSSy.  he was
aDtoHIatiH, neKeI aSSowinJ hisstaCC to disaJIee with him, bDt otheIwise not
DnUSeasant so SonJ asemUSoyees IeHoJniMed that he was boss.  a HaIdinaS
aItiHSe oC Caith withhim was that a weSS-eYeHDted saSes HamUaiJn HoDSd UDt
aHIoss aSmostanythinJ.
```

3.4 Iterația 4: Reconstituirea numelor proprii istorice (`Friedman`, `Yardley`)

În această etapă, textul devine mult mai ușor de citit și putem identifica numele proprii de persoane:

1. În sintagma `"the JIeatest is wiSSiam"`, cuvântul `wiSSiam` este fără îndoială prenumele `"William"`, ceea ce ne dă $\texttt{S} \to \texttt{l}$.

2. Numele din continuare `CIedeIiHL` are structura `"C-r-e-d-e-r-i-c-k"`, corespunzând numelui `"Frederick"`, de unde obținem: $\texttt{C} \to \texttt{f}$, $\texttt{I} \to \texttt{r}$, $\texttt{H} \to \texttt{c}$ și $\texttt{L} \to \texttt{k}$.

3. Numele de familie `CIiedman` devine `"Friedman"`, făcând referire la cunoscutul criptolog **William Frederick Friedman**.

4. La începutul textului, numele `"heIbeIt yaIdSey"` devine `"herbert yardley"` (**Herbert Yardley**), confirmând literele deduse anterior.

Am completat aceste litere în instrument (`S` → `l`, `C` → `f`, `I` → `r`, `H` → `c`, `L` → `k`) și am apăsat *"Make Substitutions"*:

```
while herbert yardley may be the best known cryUtoloJist, Dncontestably the
Jreatest is william frederick friedman.  Dnlike hiscontemUorary, his
eminence is dDe most emUhatically to what he did.  indeed, two more
dissimilar men in a sinJle field can scarcely beimaJined.  where yardley
was rabelaisian, oDtJoinJ, sDUerficial, free andeasy with the details of a
Jood story, and eKer ready for the main chance, friedman tended toward
introKersion, deUth of stDdy, Uersonal secDrity, timidity, dedication, and
accDracy, nicety, and Kalidity of work.  desUitethe relatiKe drabness of
these Uersonal traits--or UerhaUs becaDse ofthem, friedman's theoretical
contribDtions and his Uractical attainmentseYceed those of any other
cryUtoloJist.  yardley's career was like anamaMinJ skyrocket that eYUlodes
in fantastic Uatterns aJainst theheaKens.  friedman's was like the sDn.  he
was born wolfe friedman on seUtember 24, 1891, in kishineK, rDssia, the
oldest son and second child of frederick and rosa friedman.  his father, a
rDmanian who sUoke eiJht lanJDaJes and worked as aninterUreter for the
rDssian Uost office, emiJrated to america in 1892, at which time his son's
name was chanJed to william.  the family settledin UittsbDrJh, where his
father manaJed a sewinJ machine aJency.  william JradDated in 1909 as one
of the ten honor stDdents in a class of300 at UittsbDrJh central hiJh
school; he then went to work as chiefclerk in the erie city iron works, a
firm that sold steam enJines.  aboDtthat time the back-to-the-farm moKement
called to city boys, and in thefall of 1910, friedman and three friends
enrolled in michiJanaJricDltDral colleJe, whose chief attraction was that
it was tDition-free.  bDt friedman soon discoKered that farminJ held little
interest for him.  he was an inKentiKe yoDnJ fellow who liked to fiY thinJs
and had writtensome science fiction for his hiJh-school UaUer; he was
raUidly cominJ tothe conclDsion that he liked science.  at the end of the
term he learnedthat tDition was also free in a scientific field allied to
aJricDltDre--Jenetics--at one of the iKy leaJDe DniKersities, cornell.  he
borrowedtrain fare and arriKed in ithaca, new york, in febrDary, 1911,
where heJot a Eob waitinJ on tables.  after commencement in febrDary of
1914, heattended JradDate school, manaJinJ to fall in loKe twice, once with
abrDnette, once with the blonde daDJhter of a moKie-hoDse owner.  whilehe
was there, a wealthy teYtile merchant, JeorJe fabyan, whomaintained
laboratories in acoDstics, chemistry, Jenetics, and cryU-toloJy (to try to
UroKe that bacon wrote shakesUeare's Ulays) on his 500-acre estate,
riKerbank, at JeneKa, illinois, decided that he needed aJeneticist to
imUroKe the Jrains and liKestock on his farm.  he aUUlied tocornell for a
"woDld-be-er," not an "as-is-er," and hired friedman, tobeJin EDne 1, 1915.
fabyan was a man of no formal edDcation bDt of intelliJence andenerJy.  he
had a Jreat desire to be "somebody," and that desiremotiKated his
sDbsidiMinJ the baconian stDdies:  Uroof of thisreKolDtionary thesis woDld
coKer its Uatron as well as its actDaldiscoKerers with Jlory.  he himself
read little, bDt he absorbed enoDJhfrom those aroDnd him to make his talk
on almost any sDbEect soDndimUressiKe--at least sDUerficially.  he was
aDtocratic, neKer allowinJ hisstaff to disaJree with him, bDt otherwise not
DnUleasant so lonJ asemUloyees recoJniMed that he was boss.  a cardinal
article of faith withhim was that a well-eYecDted sales camUaiJn coDld UDt
across almostanythinJ.
```

3.5 Iterația 5: Decriptarea integrală a mesajului

Textul este acum lizibil în proporție de peste 90%, iar literele rămase se pot deduce direct din cuvintele parțiale:

1. În cuvântul `"Dncontestably"`, prefixul negativ este clar `"un-"`, deci `D` → `u` (*"uncontestably"*).
2. În cuvântul `"Jreatest"`, lipsește prima literă: `J` → `g` (*"greatest"*).
3. În expresia `"eKer ready for the main chance"`, recunoaștem adverbul `"ever"`, deci `K` → `v` (*"ever ready"*).
4. În cuvintele `"cryUtoloJist"` și `"Uractical"`, este clar vorba despre litera `p`, deci `U` → `p` (*"cryptologist"*, *"practical"*).
5. În expresia `"liked to cxy things"`, recunoaștem verbul `"fix"`, deci `Y` → `x` (*"fix things"*).
6. În data `"Edgv 1, 1915"`, recunoaștem luna iunie, deci `E` → `j` (*"June 1, 1915"*).
7. În cuvântul `"tgtztmxgj"` → `"amaMinJ"`, recunoaștem adjectivul `"amazing"`, deci `M` → `z`.
8. Litera `B` are 0 apariții în text deoarece litera clară `"q"` nu apare în acest pasaj. Prin eliminare (fiind singura literă rămasă din alfabetul de 26 de caractere), deducem că `B` corespunde lui `q`.

După completarea tuturor literelor în tabelul instrumentului și apăsarea butonului *"Make Substitutions"*, obținem textul descifrat complet:

```
while herbert yardley may be the best known cryptologist, uncontestably the
greatest is william frederick friedman.  unlike hiscontemporary, his
eminence is due most emphatically to what he did.  indeed, two more
dissimilar men in a single field can scarcely beimagined.  where yardley
was rabelaisian, outgoing, superficial, free andeasy with the details of a
good story, and ever ready for the main chance, friedman tended toward
introversion, depth of study, personal security, timidity, dedication, and
accuracy, nicety, and validity of work.  despitethe relative drabness of
these personal traits--or perhaps because ofthem, friedman's theoretical
contributions and his practical attainmentsexceed those of any other
cryptologist.  yardley's career was like anamazing skyrocket that explodes
in fantastic patterns against theheavens.  friedman's was like the sun.  he
was born wolfe friedman on september 24, 1891, in kishinev, russia, the
oldest son and second child of frederick and rosa friedman.  his father, a
rumanian who spoke eight languages and worked as aninterpreter for the
russian post office, emigrated to america in 1892, at which time his son's
name was changed to william.  the family settledin pittsburgh, where his
father managed a sewing machine agency.  william graduated in 1909 as one
of the ten honor students in a class of300 at pittsburgh central high
school; he then went to work as chiefclerk in the erie city iron works, a
firm that sold steam engines.  aboutthat time the back-to-the-farm movement
called to city boys, and in thefall of 1910, friedman and three friends
enrolled in michiganagricultural college, whose chief attraction was that
it was tuition-free.  but friedman soon discovered that farming held little
interest for him.  he was an inventive young fellow who liked to fix things
and had writtensome science fiction for his high-school paper; he was
rapidly coming tothe conclusion that he liked science.  at the end of the
term he learnedthat tuition was also free in a scientific field allied to
agriculture--genetics--at one of the ivy league universities, cornell.  he
```

```
borrowedtrain fare and arrived in ithaca, new york, in february, 1911,
where hegot a job waiting on tables.  after commencement in february of
1914, heattended graduate school, managing to fall in love twice, once with
abrunette, once with the blonde daughter of a movie-house owner.  whilehe
was there, a wealthy textile merchant, george fabyan, whomaintained
laboratories in acoustics, chemistry, genetics, and cryp-tology (to try to
prove that bacon wrote shakespeare's plays) on his 500-acre estate,
riverbank, at geneva, illinois, decided that he needed ageneticist to
improve the grains and livestock on his farm.  he applied tocornell for a
"would-be-er," not an "as-is-er," and hired friedman, tobegin june 1, 1915.
fabyan was a man of no formal education but of intelligence andenergy.  he
had a great desire to be "somebody," and that desiremotivated his
subsidizing the baconian studies:  proof of thisrevolutionary thesis would
cover its patron as well as its actualdiscoverers with glory.  he himself
read little, but he absorbed enoughfrom those around him to make his talk
on almost any subject soundimpressive--at least superficially.  he was
autocratic, never allowing hisstaff to disagree with him, but otherwise not
unpleasant so long asemployees recognized that he was boss.  a cardinal
article of faith withhim was that a well-executed sales campaign could put
across almostanything.
```

Formatat cu litere mari adecvate și așezat pe paragrafe, textul decriptat se citește astfel:

> While Herbert Yardley may be the best known cryptologist, uncontestably the greatest is William Frederick Friedman. Unlike his contemporary, his eminence is due most emphatically to what he did. Indeed, two more dissimilar men in a single field can scarcely be imagined. Where Yardley was Rabelaisian, outgoing, superficial, free and easy with the details of a good story, and ever ready for the main chance, Friedman tended toward introversion, depth of study, personal security, timidity, dedication, and accuracy, nicety, and validity of work. Despite the relative drabness of these personal traits—or perhaps because of them, Friedman's theoretical contributions and his practical attainments exceed those of any other cryptologist. Yardley's career was like an amazing skyrocket that explodes in fantastic patterns against the heavens. Friedman's was like the sun.
>
> He was born Wolfe Friedman on September 24, 1891, in Kishinev, Russia, the oldest son and second child of Frederick and Rosa Friedman. His father, a Rumanian who spoke eight languages and worked as an interpreter for the Russian Post Office, emigrated to America in 1892, at which time his son's name was changed to William. The family settled in Pittsburgh, where his father managed a sewing machine agency. William graduated in 1909 as one of the ten honor students in a class of 300 at Pittsburgh Central High School; he then went to work as chief clerk in the Erie City Iron Works, a firm that sold steam engines. About that time the back-to-the-farm movement called to city boys, and in the fall of 1910, Friedman and three friends enrolled in Michigan Agricultural College, whose chief attraction was that it was tuition-free.
>
> But Friedman soon discovered that farming held little interest for him. He was an inventive young fellow who liked to fix things and had written some science fiction for his high-school paper; he was rapidly coming to the conclusion that he liked science. At the end of the term he learned that tuition was also free in a scientific field allied to agriculture—genetics—at one of the Ivy League universities, Cornell. He borrowed train fare and arrived in Ithaca, New York, in February, 1911, where he got a job waiting on tables. After commencement in February of 1914, he attended graduate school, managing to fall in love twice, once with a brunette, once with the blonde daughter of a movie-house owner.
>
> While he was there, a wealthy textile merchant, George Fabyan, who maintained laboratories in acoustics, chemistry, genetics, and cryptology (to try to prove that Bacon

> wrote Shakespeare's plays) on his 500-acre estate, Riverbank, at Geneva, Illinois, decided that he needed a geneticist to improve the grains and livestock on his farm. He applied to Cornell for a "would-be-er," not an "as-is-er," and hired Friedman, to begin June 1, 1915.
>
> Fabyan was a man of no formal education but of intelligence and energy. He had a great desire to be "somebody," and that desire motivated his subsidizing the Baconian studies: proof of this revolutionary thesis would cover its patron as well as its actual discoverers with glory. He himself read little, but he absorbed enough from those around him to make his talk on almost any subject sound impressive—at least superficially. He was autocratic, never allowing his staff to disagree with him, but otherwise not unpleasant so long as employees recognized that he was boss. A cardinal article of faith with him was that a well-executed sales campaign could put across almost anything.

Acesta este un extras din lucrarea istorică *"The Codebreakers: The Story of Secret Writing"* de David Kahn (1967), Capitolul 12 (*"Two Americans"*).

4 Reconstituirea Alfabetului Cheii (Tabelul 2.5)

De asemenea, acum putem recupera cheia folosită în criptare prin alăturarea alfabetelor textului clar și a mesajului criptat interceptat (Tabelul 2.5). Acest lucru este util dacă avem și alte mesaje interceptate de la aceeași sursă, deoarece este probabil să folosească aceeași cheie (sau o rotație a două sau trei chei).

Table 4: Alfabetul reconstituit al mesajului criptat (Tabelul 2.5).

Plaintext Alphabet	a	b	c	d	e	f	g	h	i	j	k	l	m
Ciphertext Alphabet	T	A	H	O	V	C	J	Q	X	E	L	S	Z
Plaintext Alphabet	n	o	p	q	r	s	t	u	v	w	x	y	z
Ciphertext Alphabet	G	N	U	B	I	P	W	D	K	R	Y	F	M

5 Concluzii

În cadrul acestei lucrări de laborator am descifrat cu succes mesajul interceptat din Varianta 22, aplicând metoda analizei de frecvență cu ajutorul instrumentului online recomandat.

Principalele concluzii ale lucrării sunt:

- **Vulnerabilitatea cifrului monoalfabetic:** Deși numărul teoretic de chei posibile este uriaș (26!), securitatea cifrului este nulă în practică dacă avem un text de lungime rezonabilă, deoarece frecvențele caracterelor din limba sursă nu sunt mascate.
- **Importanța contextului lingvistic:** O decriptare pur mecanică bazată doar pe frecvența literelor individuale nu este suficientă din cauza variațiilor locale din text. Spargerea eficientă a mesajului a fost posibilă prin combinarea unigramelor cu identificarea trigrafelor dominante (`WQV` $\rightarrow$ `the`), a cuvintelor scurte (`to`, `that`, `is`, `in`, `and`) și a numelor proprii (`William Frederick Friedman`, `Yardley`).
- **Eficiența procesului iterativ:** Pe măsură ce primele litere cheie au fost găsite, restul cuvintelor au devenit mult mai ușor de recunoscut, confirmând faptul că efortul principal constă în identificarea corectă a nucleului inițial.

6 Depozit Cod Sursă

Fișierele raportului sunt organizate în depozitul proiectului:

Link Depozit GitHub:
https://github.com/Lucariowu/cslabs/tree/main/lab2